

★ RS003 Week 10 — English Worksheet

Topic: Feedback When You Hit the Wall · **Course:** Pygame Turret · **Time:** about 45 minutes

This week the boundary **reacts**. When the aim slams into a wall, the line flashes red and the screen shakes for a moment. You learn to use **timer variables** that count down frame by frame, and **random** to make a shake.

Words to know: **timer, flash, shake, offset, random**

1 · Predict 🧠

Read each piece of code. Before you run it, write what you think will happen.

```
flash_timer = 0
if gun_tip_x > MAX_X:
    gun_tip_x = MAX_X
    flash_timer = 10
```

When the gun hits the right wall, **flash_timer** is set to 10. What might 10 mean here?


```
if flash_timer > 0:
    flash_timer = flash_timer - 1
```

This runs every loop. How long does **flash_timer** stay above 0 after being set to 10?


```
shake_x = random.randint(-5, 5) if shake_timer > 0 else 0
pygame.draw.rect(screen, GREEN, (CENTER_X - 30 + shake_x, HEIGHT - 80, 60, 40))
```

While the shake timer is running, `shake_x` is a small random number. What does adding it to every position do?

2 · Run It

 **Type the example, run it, and slam the wall**

```

import pygame
import random
pygame.init()

WIDTH, HEIGHT = 800, 600
CENTER_X = WIDTH // 2
screen = pygame.display.set_mode((WIDTH, HEIGHT))
clock = pygame.time.Clock()

MIN_X = 50
MAX_X = WIDTH - 50
gun_tip_x = CENTER_X
SPEED = 6
flash_timer = 0
shake_timer = 0

running = True
while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False

    keys = pygame.key.get_pressed()
    if keys[pygame.K_LEFT]:
        gun_tip_x -= SPEED
    if keys[pygame.K_RIGHT]:
        gun_tip_x += SPEED

    if gun_tip_x < MIN_X:
        gun_tip_x = MIN_X
        flash_timer = 10
        shake_timer = 10
    if gun_tip_x > MAX_X:
        gun_tip_x = MAX_X
        flash_timer = 10
        shake_timer = 10

    shake_x = random.randint(-5, 5) if shake_timer > 0 else 0
    shake_y = random.randint(-5, 5) if shake_timer > 0 else 0

    screen.fill((135, 206, 235))
    line_color = (255, 50, 50) if flash_timer > 0 else (180, 180, 180)
    line_width = 6 if flash_timer > 0 else 2
    pygame.draw.line(screen, line_color, (MIN_X + shake_x, 0), (MIN_X + shake_x,
HEIGHT), line_width)

```

```
pygame.draw.line(screen, line_color, (MAX_X + shake_x, 0), (MAX_X + shake_x,
HEIGHT), line_width)
pygame.draw.rect(screen, (0, 100, 0), (CENTER_X - 30 + shake_x, HEIGHT - 80 +
shake_y, 60, 40))
pygame.draw.line(screen, (80, 80, 80), (CENTER_X + shake_x, HEIGHT - 80 +
shake_y), (gun_tip_x + shake_x, HEIGHT - 200 + shake_y), 10)
pygame.display.flip()

if flash_timer > 0:
    flash_timer = flash_timer - 1
if shake_timer > 0:
    shake_timer = shake_timer - 1

clock.tick(60)

pygame.quit()
```

Slam into a wall. Describe what flashes and what shakes, and how long it lasts:

3 · Spot the Bug 🐛

Each block below was meant to do something but is broken. Fix it, then explain why the original was wrong.

Bug A — The wall should flash for a moment then go back to grey, but once it flashes it stays red forever.

```
if gun_tip_x > MAX_X:
    gun_tip_x = MAX_X
    flash_timer = 10
```

Hint: the timer is set but never counts down.

Write the fixed code (show where the timer goes down):

Why was it wrong? Why does your fix work?

Bug B — The screen should only shake when you hit a wall, but it shakes all the time even in the middle.

```
shake_x = random.randint(-5, 5)
```

Hint: only shake while the shake timer is running.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The shake should move everything together, but right now the barrel jumps around while the base stays still, so they come apart.

```
pygame.draw.rect(screen, GREEN, (CENTER_X - 30, HEIGHT - 80, 60, 40))
pygame.draw.line(screen, GRAY, (CENTER_X + shake_x, HEIGHT - 80), (gun_tip_x +
shake_x, HEIGHT - 200), 10)
```

Hint: the base does not have the shake offset added.

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Modify It

Tune the shake strength and length

Change `random.randint(-5, 5)` to a bigger range, and change the timer from 10 to a bigger or smaller number.

Write the shake range and timer length you chose, and how it feels:

Design your own hit feedback

Replace or add to the flash with your own idea — a flash of the whole screen, a colour change on the barrel, anything visual.

Describe your feedback idea and the timer that controls it:

Hint: any feedback works the same way — set a timer to a number on the hit, count it down each loop, and do something special while it is above 0.

5 · Make It 📷

Build a turret that reacts to walls

Build a program where:

1. the aim is clamped to the screen,
2. hitting a wall flashes the boundary,
3. hitting a wall shakes the scene for a moment,
4. everything shakes together.

Send a **video** showing a wall hit, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I made a timer that ...

When the gun hits a wall, I set the timer to ...

Each loop the timer goes down by ...

The shake uses random numbers to ...

One tricky moment was when ...

If I had more time, I would ...

6 · Record Your Walkthrough 🎥

Take a video showing a wall hit and the reaction. Talk through it like you are teaching someone. Try to use these words: **timer, flash, shake, offset, random.**

1. Aim into a wall and show the flash and shake.
2. Read your timer line out loud and say what 10 means.
3. Point at where the timer counts down.
4. Show your own feedback idea.

Write what you will say in your video. Plan it here before you record.

Submit ✓

Send this worksheet + your wall-hit video to teacher on KakaoTalk.